

Data Science and Engineering

Machine Learning in Health

Lab 3. Multi-view VAEs

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Lab III: Multi-View VAE

Code and implementation can be found here: [\[ML_in_Healthcare\]_Lab3 Final.ipynb](#)

Dataset



We built the paired dataset using the class `PairedMNISTSVHN(Dataset)` with the corresponding label by creating buckets with each labelled image and randomizing the pairings. This generated a training set of 56,068 paired images and 520 for the test set.

To plot with `imshow` we apply a squeeze onto MNIST sample to remove grayscale channel and permute the position of SVHN channel to the last position.

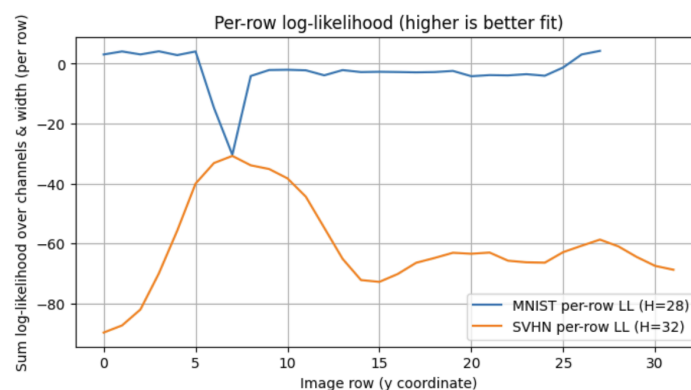
Multi-view VAE Architecture

Encoder-Decoder

For the MNIST dataset, which is simpler and only contains one channel, was a convolutional model of five convolutional layers. On the other hand, for the SVHN dataset five layers were also implemented but with 256 channels in the latent space instead of 112 for the MNIST dataset.

Loss Function

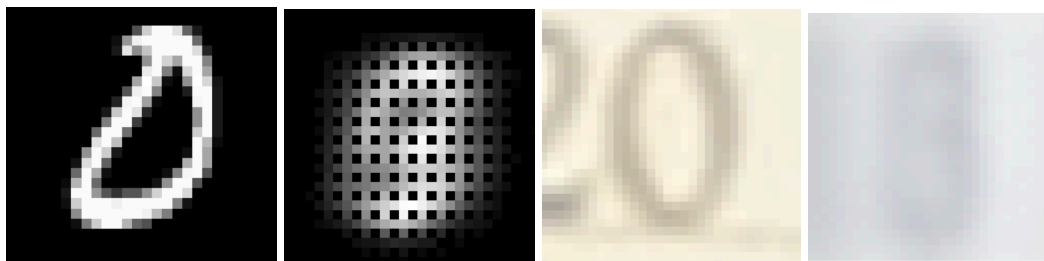
For the loss function we use the calculation of the accumulated Gaussian log-likelihood for all datapoints in the data and plot by row. Since MNIST and SVHN have different dimensions we normalize.



Latent Space Visualization and Reconstruction

After training the Multi-View VAE, our goal was to analyze whether the model successfully learned a shared latent representation that captures semantic similarity between MNIST and SVHN images. Since the latent dimension is 10-dimensional and cannot be directly visualized, we performed a t-SNE projection to reduce the latent vectors into a 2-dimensional space.

To extract the latent embeddings, we implemented the `MultiViewLatent` module. For each MNIST–SVHN pair in the test dataset, we obtained the fused latent mean vector produced by the Product-of-Experts combination of the two encoders. This fused mean represents the center of the approximate posterior distribution and serves as a stable point for visual comparison across samples. We then applied t-SNE with standard hyperparameters (perplexity = 30, learning rate = 200) to embed all latent vectors into a two-dimensional map.



Interpretation of the t-SNE Plot

The resulting 2-D embedding shows several important properties that show the proper working of our VAE. Firstly, the digit clusters are clearly separated. Points belonging to the same label (0–9) form consistent and compact regions in the latent space, which preserves semantic structure. The clusters are smooth as well, with similar digits closer to each other in the latent map; this points to a meaningful geometry learned within the neural network.

Aside from that, MNIST and SVHN samples with the same label map to the same region even though they are different in number of channels and dimensions. Thanks to the Product-of-Experts fusion, this causes both modalities to converge towards a joint latent cluster for each digit.

Conclusion

This lab was an insightful exercise into VAEs, even expanding into multi-view VAEs, and the way they are trained and evaluated. Through the use of simple, familiar datasets such as MNIST and SVHN the difficult concept became easier to break down and understand.